

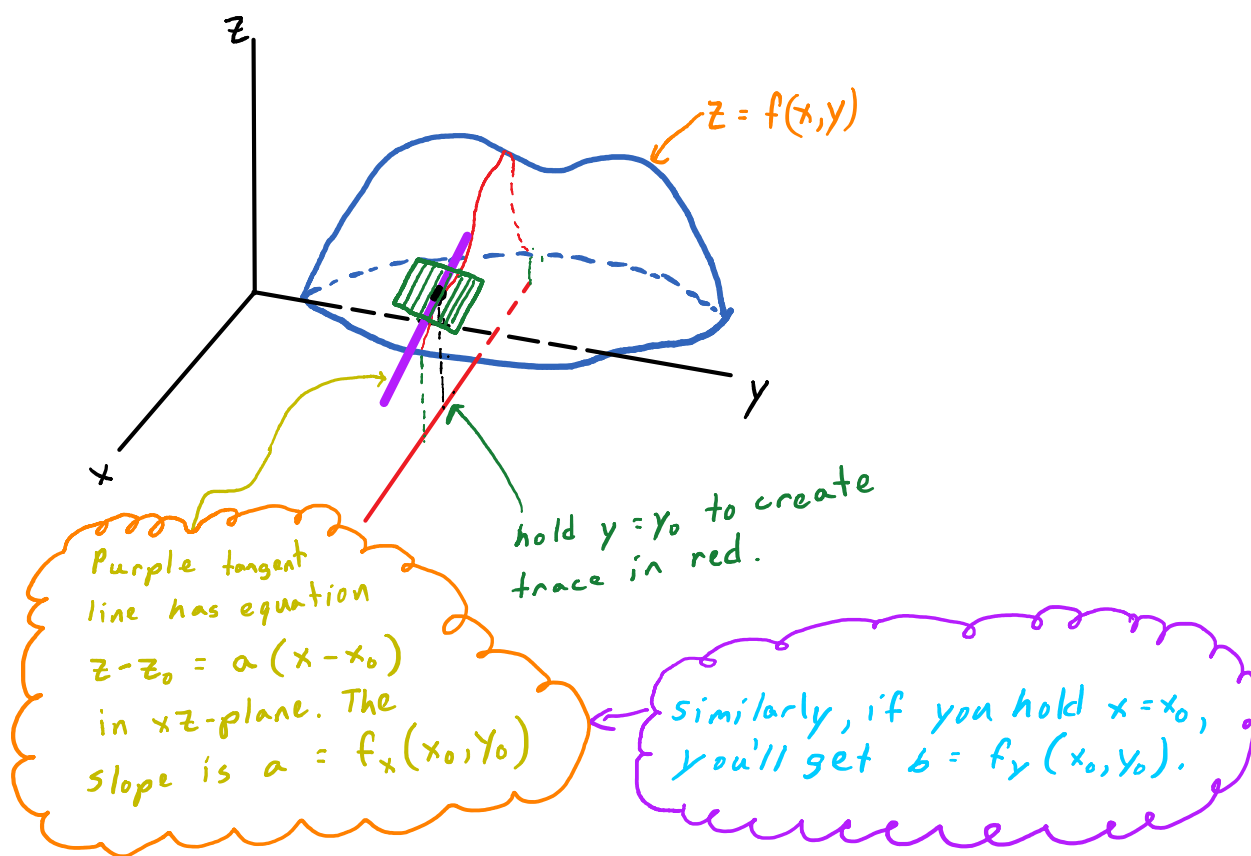
Tangent Planes and Linear Approximations

Goals:

1. To find the tangent plane to a surface at a given point.
2. To calculate the total differential of a multivariable function.
3. To determine whether or not a function is differentiable.

Derivation of the Tangent Plane to a Surface at a Point

Note that the tangent plane to $z = f(x, y)$ at (x_0, y_0, z_0) has equation of the form $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$ or $\underbrace{z - z_0 = a(x - x_0) + b(y - y_0)}_{\text{solve for this}}.$



So, the equation of the tangent plane is ...

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(y - y_0)$$

(ex) a) Find the equation of the tangent plane to

$$f(x, y) = x^2 - 2xy + y^2 \text{ at } (1, 2, 1).$$

$$f_x(x, y) = 2x - 2y \rightarrow f_x(1, 2) = 2 - 4 = -2$$

$$f_y(x, y) = -2x + 2y \rightarrow f_y(1, 2) = -2 + 4 = 2$$

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

$$z - 1 = -2(x - 1) + 2(y - 2)$$

b) Find a linearization, $L(x, y)$, for $f(x, y)$ at $(1, 2, 1)$.
z from

$$L(x, y) = -2(x - 1) + 2(y - 2) + 1$$

$$= -2x + 2 + 2y - 4 + 1$$

$$= -2x + 2y - 1$$

c) Use $L(x, y)$ to estimate $f(0.95, 2.02)$

$$L(0.95, 2.02) = -2(0.95) + 2(2.02) - 1$$

$$= 1.14$$

$$[\text{Turns out } f(0.95, 2.02) \approx 1.1449]$$

②x Consider the tangent plane equation:

$$z = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) + z_0.$$

Let $x = x_0 + \Delta x$ and $y = y_0 + \Delta y$ and discuss what the result represents geometrically.

$$\begin{aligned} z &= f_x(x_0, y_0)((x_0 + \Delta x) - x_0) + f_y(x_0, y_0)((y_0 + \Delta y) - y_0) + z_0 \\ &= f_x(x_0, y_0) \Delta x + f_y(x_0, y_0) \Delta y + z_0 \end{aligned}$$

change in height of the tangent plane from inputs of (x_0, y_0) to inputs of $(x_0 + \Delta x, y_0 + \Delta y)$

Definitions

① $\underbrace{dz}_{\text{total differential}} = f_x(x, y) \underbrace{dx}_{dx = \Delta x} + f_y(x, y) \underbrace{dy}_{dy = \Delta y}$

Recall ① from calc I:

$$\underbrace{dy = f'(x) dx}_{\text{differential for } y = f(x)}$$

for a particular x_0 , this is the change in tangent line height from x_0 to $x_0 + \Delta x$.

② $\Delta y \approx dy$ where $\Delta y = f(x_0 + \Delta x) - f(x_0)$,

f differentiable and Δx "small."

$[\Delta y = \text{change in height of } y = f(x_0) \text{ from } x_0 \text{ to } x_0 + \Delta x]$

$$(2) \quad \Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$$

change in height
of f from (x_0, y_0)
to $(x_0 + \Delta x, y_0 + \Delta y)$

Note: For "well-behaved" functions and for "small" $\Delta x, \Delta y$, $dz \approx \Delta z$ when going from inputs (x_0, y_0) to $(x_0 + \Delta x, y_0 + \Delta y)$.

(ex) Let $z = \sin(2x + 3y)$. Use dz to estimate Δz from $(-3, 2)$ to $(-2.95, 2.04)$

$$dz = f_x(x_0, y_0) \Delta x + f_y(x_0, y_0) \Delta y$$

$$f_x(x, y) = 2 \cos(2x + 3y) \rightarrow f_x(-3, 2) = 2$$

$$f_y(x, y) = 3 \cos(2x + 3y) \rightarrow f_y(-3, 2) = 3$$

$$dz = 2(0.05) + 3(0.04) = \boxed{0.22}$$

Note: $\Delta z \approx 0.2182$

Note: ① In calc I, differentiability $\Rightarrow$ continuity.

② Problem: partials exist $\nRightarrow$ f is continuous

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↑ see (ex) below

[We want a definition of differentiability for $z = f(x, y)$ such that f differentiable $\Rightarrow$ f is continuous.]

③ solution: come up with a definition of a differentiable function that ensures $\Delta z \approx dz$ for "small" input changes.

Definition: $z = f(x, y)$ is differentiable at (x_0, y_0) as long as Δz can be written as ...

$$\Delta z = \underbrace{f_x(x_0, y_0) \Delta x + f_y(x_0, y_0) \Delta y}_{dz|_{(x_0, y_0)}} + \epsilon_1 \Delta x + \epsilon_2 \Delta y$$

where $\epsilon_1, \epsilon_2 \rightarrow 0$ as $(\Delta x, \Delta y) \rightarrow (0, 0)$.

Theorem 1: $z = f(x, y)$ differentiable at (x_0, y_0)

$\Rightarrow f$ is continuous (x_0, y_0) .

Proof: uses the above def to show

$$\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = f(x_0, y_0)$$

means f is continuous at (x_0, y_0)

Theorem 2: f_x, f_y continuous on D , then f is differentiable on D .

Partials exist but f not differentiable at $(0,0)$!

(ex) Let
$$f(x,y) = \begin{cases} \frac{2xy}{2x^2 + y^2} & , (x,y) \neq (0,0) \\ 0 & , (x,y) = (0,0) \end{cases}$$

Find the partials $f_x(0,0)$ and $f_y(0,0)$, if they exist.

$$f_x(0,0) = \lim_{\Delta x \rightarrow 0} \frac{f(0+\Delta x, 0) - f(0,0)}{\Delta x} = \frac{2 \cdot \Delta x \cdot 0}{2(\Delta x)^2} = 0$$

$$f_y(0,0) = \lim_{\Delta y \rightarrow 0} \frac{f(0, 0+\Delta y) - f(0,0)}{\Delta y} = \frac{2 \cdot 0 \cdot \Delta y}{(\Delta y)^2} = 0$$

But $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$ DNE! So, f is not

continuous at $(0,0)$ but its partials exist.

[i.e. partials exist $\nRightarrow$ f differentiable].

We showed this in 14.2

So, just because the partials exist at a point does not mean that the function is well-behaved at that point.

f behaving badly
around input $(0,0)$

